Universal Fault-Tolerant Computation on Three-Dimensional Layer Codes at Optimal Distance

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Three-dimensional layer codes combine local stabilizer checks with high storage density, but computation also requires local access to their logical operators. We give an explicit local realization of the Williamson–Baspin layer code for any CSS input whose checks have weight at least two, and prove that it encodes exactly the input logical qubits, that every pure-X logical class has a representative on a single lattice layer, at every height of the block, and that every pure-Z class has one in a single transverse plane, at every position. A running (prefix) parity along each check sheet gives these representatives. The slabs support logical Pauli measurements by nearest-neighbor three-dimensional grid gauging. A constant number of ancillas per grid edge preserves the full code distance, whereas a measurement layer of bounded thickness provably cannot: a thin interface contains a wall of few edges that a few faults can cross. Circuit-level simulations on explicit layer-code blocks confirm both statements; as the blocks grow, failures through the grid fall and failures through a one-row interface rise. We then observe these distance mechanisms on IQM and IBM superconducting processors, with three-qubit blocks and 6.9 million samples. Widening the interface from one edge to two lowers the error of a joint measurement 35-fold on IQM (from 2.4% to 0.070%, with no failure in 11,529 samples at three edges) and 9.7-fold on IBM, and two parallel ancillas on one edge lower it 4.6-fold on IQM. Repeated rounds protect the reported bit as a calibrated model predicts when every round uses fresh ancillas. When ancillas are measured and reset instead, errors that survive the reset, most likely leakage, leave the three-round error 68 times above the calibrated Pauli model, a failure mode outside local-stochastic noise. Lifting the input distance through the slabs shows that the realization has the optimal distance $\Theta(N^2)$ for good inputs. Combining the measurements with encoded resource transfer, distillation, and a local detector schedule gives a universal fault-tolerant architecture under local-stochastic Pauli noise. At fixed noise below an existential threshold, a depth-T circuit on W logical sites in a bounded-aspect box uses $O(Wf + f^{3/2})$ physical qubits and $O(Tf^{3/2}\Psi) \subseteq O(Tf^{5.41})$ depth, where $f = \Theta(\log(WT/\varepsilon))$ and $\Psi = O(f^{\log_2 15})$ is the distillation depth factor; a concrete estimate places the storage advantage at distances of order a thousand. We also derive a fragmentation correction for general regions and port-packing bounds saturated by balanced layer codes.

I. INTRODUCTION

Quantum error-correcting codes must protect stored information and make that information accessible to computation. These demands can pull in different directions: a code with favorable storage parameters may still require costly logical measurements or physical connections that are difficult to realize. Three-dimensional layer codes provide a natural setting in which to address this problem. The construction of Williamson and Baspin embeds a CSS code into a local network of surface-code sheets and defects [1]. For balanced good low-density parity-check inputs of length N, the resulting blocks have

$$n = \Theta(N^3), \qquad K = \Theta(N), \qquad d = \Theta(N^2), \quad (1)$$

with bounded-weight, geometrically local stabilizers. Here n is the physical block size, K its number of logical qubits, and d its distance.

The central result of this work is a geometric interface for processing that stored information. We give an explicit lattice realization of the layer code with complete check matrices, and show that every pure-X logical class has a representative on a single layer of the lattice and every pure-Z class one in a single transverse plane. The layer can be placed at any height, and the plane at any position, independently of N and of the chosen logical operator. This converts an algebraic choice of logical Pauli into a spatially localized measurement port.

We use this interface to construct universal fault-tolerant computation on the native three-dimensional nearest-neighbor lattice. The argument has three steps. A prefix-parity construction first produces the slab representatives. Relative expansion of a cubic grid then gives a local gauging measurement with distance $\Omega(d)$. Finally, repeated extraction, padded composition, and encoded resource transfer give a complete circuit under local-stochastic Pauli noise. Figure 1 summarizes the architecture.

Logical measurement by gauging, surgery, and homological constructions has become a general route to processing quantum LDPC codes [2–6]. Adapters, extractors, and universal architectures address complementary questions of connectivity, parallelism, and resource overhead [7–10]. Here the organizing constraint is three-dimensional Euclidean locality: logical access, resource delivery, and syndrome extraction use the same nearest-neighbor geometry. The layer-code slab theorem supplies the missing connection between the memory construction and these processing primitives.

Locality also fixes the achievable overhead. In three dimensions a local stabilizer code obeys $Kd = O(n)$ [11], so $\Theta(d)$ physical qubits per logical qubit is the best possible storage cost, and layer codes attain it. Surface-code

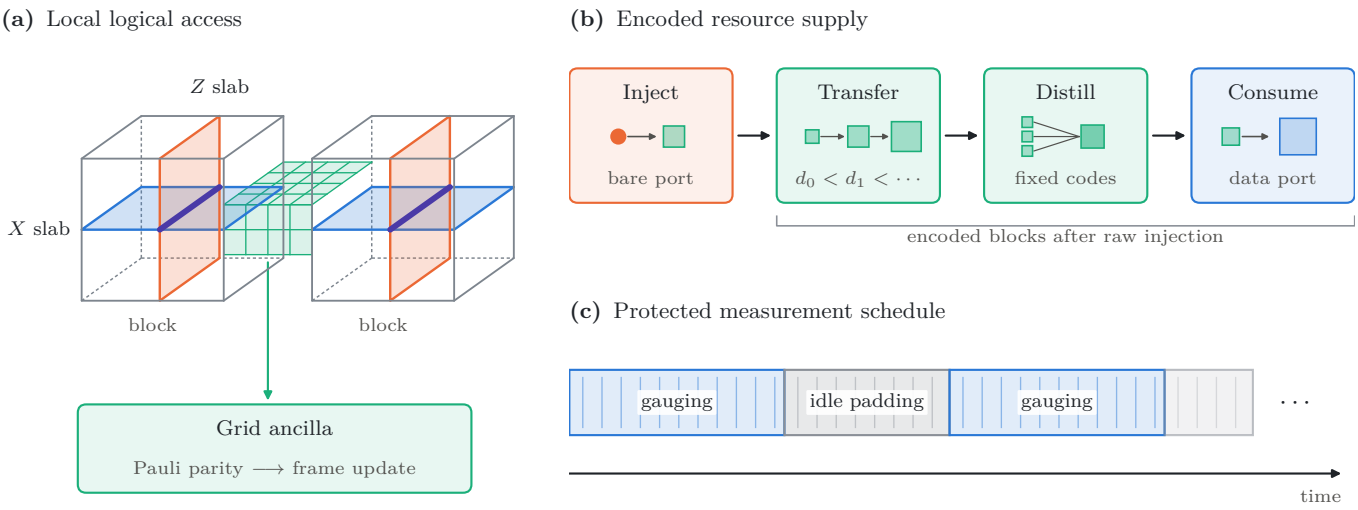


FIG. 1. Geometry-preserving processing. (a) A layer-code block exposes transverse slab ports for pure-X and pure-Z logical operators. A local grid ancilla measures the selected Pauli; neighboring blocks use the same interface for joint measurements. (b) A raw resource is injected into an encoded block, transferred through blocks of increasing distance, distilled, and consumed by the data path. (c) Repeated extraction and idle padding connect successive active measurements. All drawings are schematic; the encoded blocks and grid ancillas occupy three-dimensional regions.

patches processed by lattice surgery use $\Theta(d^2)$ qubits per logical qubit [12, 13], and three-dimensional surface and color codes trade larger volume for transversal gates [14]. Local simulations of LDPC codes [15], local fault-tolerance constructions [16], wire codes [17], and planar logical measurements [18] address related geometric questions, while constant-overhead schemes [19–21] and LDPC magic-state injection [22, 23] rely on nonlocal connectivity. The architecture below keeps the $\Theta(d)$-per-logical storage cost during computation, up to the cost of one encoded cell.

The distance mechanisms behind the gauging measurements already act in circuits small enough for present processors, so we also observe them on hardware (Sec. VII and Fig. 6). On IQM and IBM superconducting processors, with three-qubit blocks in place of layer-code blocks, a joint measurement through an interface of one edge fails 35 times more often than through an interface of two edges on IQM, and 9.7 times more often on IBM, as the cut obstruction predicts; two parallel ancillas on the single edge lower the error 4.6-fold on IQM; and repeated rounds protect the reported bit. The runs also expose a limit of the noise model. On IBM, ancilla errors that survive a reset, most likely leakage, repeat the same wrong outcome in every round and leave the three-round error 68 times above a calibrated Pauli model. Such correlated faults lie outside the local-stochastic model of our theorem, so an implementation of its schedule would need leakage reduction.

The main text develops the construction and the resource accounting, with proof sketches at each step, and reports a complete circuit-level simulation of one two-block logical measurement on explicit layer-code blocks and the hardware demonstrations of its distance mechanisms. Complete proofs, cell coordinates, detector templates, and numerical controls appear in the Supplemental Material [24]. We also show that the same slab mobility saturates a logical-port packing bound and restricts shallow code-preserving unitaries. These structural consequences are independent of the fault-tolerance proof.

II. LOGICAL OPERATORS IN CONSTANT-THICKNESS SLABS

A. The layer code as glued sheets

A layer code is assembled from surface-code sheets [1]. Each qubit q of the input CSS code becomes a *data sheet*, the plane $y = Y_q$; each input Z check becomes a Z-check sheet, a plane $x = \text{const}$; and each input X check becomes an X-check sheet, a plane $z = \text{const}$. Every sheet is an ordinary surface code on a rectangle of the cubic lattice, so it carries string operators. A string of X operators that runs from one side of a sheet to the other commutes with all checks of that sheet; a string that stops inside the sheet leaves a violated check, a *defect*, at its end. A Z-check sheet meets the data sheets of the qubits in its check along vertical lines, the *junctions*. Ordered along the check, these are its first, middle, and last junctions. The X-check sheets are attached in the same way, with X and Z exchanged.

A junction is a rule that says which strings may end on the line without leaving a defect. For the X strings used below only their presence matters. At a first or last junction the Z-check sheet ends on the data sheet, and a string may pass from one sheet to the other: the check-sheet string is present exactly when the data string is. At

a middle junction three strings meet the line: the data string, which crosses it, and the check-sheet strings on its two sides. They leave no defect exactly when an even number of the three is present. Writing 1 for a present string, and listing the check-sheet strings first (left, then right) and the data string last,

$$H_{\text{end}} = \begin{bmatrix} 1 & 1 \end{bmatrix}, \qquad \ker H_{\text{end}} = \{00, 11\},$$
$$H_{\text{mid}} = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}, \quad \ker H_{\text{mid}} = \{000, 110, 101, 011\}. \quad (2)$$

Williamson and Baspin state these rules as sets of condensed anyons, that is, as the combinations of string endpoints that disappear at the line; Eq. (2) is the same information written for X strings (Supplemental Sec. SI B). The word 110 is the important one. It lets a string on the check sheet pass a data sheet without any string on that data sheet. If a middle junction allowed only 000 and 111, every check-sheet string would drag a string onto each data sheet it passes, and a general logical operator could not be built; such a lattice in fact fails to encode the input qubits (Supplemental Sec. SII).

B. An explicit lattice for the junctions

We build every junction on the lattice (Supplemental Sec. SI A). Sheets own distinct qubits, and each junction rule adds one qubit of one sheet to one check of the other. At a middle junction the Z-check-sheet face beside the line gains the data sheet's edge on the line. That face then contains three qubits of interest: the check-sheet edge on the line, where the string from one side ends; the next check-sheet edge, where the string on the other side begins; and the added data edge. Its check is therefore exactly the middle rule of Eq. (2). The data stars on the line gain the check sheet's edges on the same side, which keeps them commuting with the modified faces. At a first or last junction the check sheet ends on the data line and shares its edges, which gives the two-bit rule.

One more repair is needed. An X-check sheet and a Z-check sheet cross along a line, and there the junction additions of the two sheets make a few stars and faces anticommute, once for each data qubit shared by the two input checks. A row of one-qubit additions along the crossing line, which we call a *wall*, removes these conflicts. It exists exactly when the shared number is even, which is the commutation of the two input checks. The slab operators below never use wall qubits, so the walls play no further role. Every check lies within unit distance of one lattice vertex and has weight at most six. Supplemental Fig. S1 draws one junction of each kind and a wall. The junctions have the condensed anyons of Ref. [1]; we do not claim that the lattice is identical to the drawings there, and all results below are proved for this explicit lattice, whose complete check matrices the source package generates.

C. Slab representatives

Theorem II.1 (Uniform slab representatives)**.** *For every input CSS code with checks of weight at least two, the explicit realization encodes exactly K logical qubits. Every pure-X logical class has a representative supported on the single layer of z-edges spanning $[z_0, z_0{+}1]$, for every height z_0 of the block, and every pure-Z class has a representative in the plane $x = x_0$, for every x_0. The heights and planes are chosen independently of the logical classes.*

The operator is built at one fixed height z_0, one input check at a time. Choose a pure-X logical $x \in \ker H_Z$ of the input code, and put an X string at height z_0 on the data sheet of every qubit with $x_q = 1$. Each such string crosses the junction of every Z-check sheet attached to its data sheet, and on its own would leave a defect there. Along one Z-check sheet the selected junctions must therefore be joined, in pairs, by strings on the check sheet at the same height; we call these check-sheet strings the *backbone*. Pairing is possible because x commutes with the input check, so the check contains an even number of selected qubits. Walk along the check from its first qubit and switch the backbone on or off at every selected junction. With ordered incidences $x_1, \ldots, x_k$, the backbone after the ith junction is present exactly when an odd number of selected qubits lies behind it, a running count that we call the *prefix parity*,

$$p_0 = 0, \qquad p_i = \sum_{h=1}^{i} x_h, \qquad p_k = 0, \qquad (3)$$

where $p_k = 0$ is the commutation $\sum_i x_i = 0$ over $\mathbb{F}_2$. At every middle junction $p_{i-1} + p_i + x_i = 0$, so none or two of the three strings are present, as Eq. (2) requires; at the ends $p_1 = x_1$ and $p_{k-1} = x_k$. No defect remains anywhere (Fig. 2). Every piece lies in the same layer of vertical edges, so the operator is one lattice layer thick, whatever the block size and whichever logical is chosen. Building it at another height gives another representative of the same class. In the explicit realization the check is exact and local: every stabilizer meets the operator in one of the forms $x_1 + p_1$, $p_{i-1} + p_i + x_i$, $p_{k-1} + x_k$ of a single incidence, or not at all.

The pure-Z operators are built in the same way in a vertical plane $x = x_0$, with backbones on the X-check sheets. An X slab operator and a Z slab operator share only one data edge for each input qubit selected by both, so they anticommute exactly when the two input logicals do. This gives K independent logical qubits. A count shows that there are no others. On its own, each data sheet would encode one qubit and each check sheet would remove one; the junctions and walls add neither qubits nor checks; and every redundancy among the lattice checks comes from a redundancy among the input checks. In formulas, each data sheet contributes $+1$ and each check sheet -1 to $n - \#X - \#Z$, and the relations among the lattice checks inject into those among the input checks.

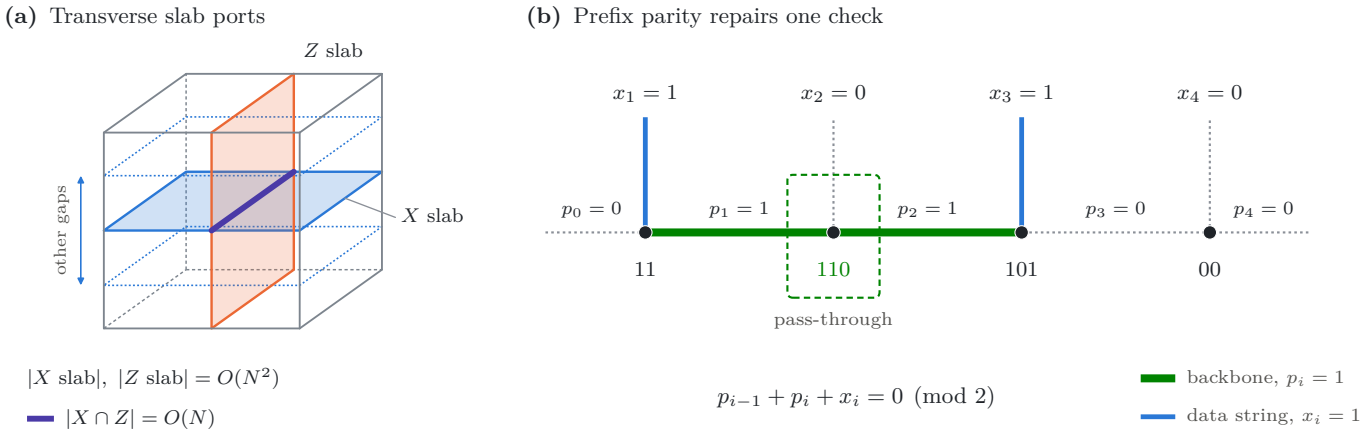


FIG. 2. Slab mobility and its local mechanism. (a) Complete X and Z ports occupy transverse single-layer slabs. Each slab contains $O(N^2)$ qubits, and their intersection contains $O(N)$ qubits. Translating to another height (or plane) gives another representative of the same logical class. (b) For the even incidence word $(1,0,1,0)$, the running parity is $(0,1,1,0,0)$. The highlighted middle cell uses 110: the backbone remains occupied although the data incidence is absent. Endpoint cells use the two-bit relation in Eq. (2).

Hence the realization encodes at most, and therefore exactly, K qubits, and the slab operators represent every pure-type class. A certificate on complete codes with up to 10,936 qubits, including the linear forms above, exhaustive negative controls, and the single-chart comparison, checks each step (Supplemental Sec. SII). The proofs are Supplemental Proposition SI.2 and Theorem SI.3.

Williamson and Baspin already obtain the layer-code logical operators by quasi-concatenating input logicals [1]. The new ingredients are an explicit realization in which this construction can be verified check by check, a single-layer thickness independent of the block size and of the chosen logical, and its uniformity over every height, which supplies the disjoint ports used below.

The same ports give the distance, by a simple idea: a logical operator must be visible in every plane that contains a complete port of the other type. Let O be a nontrivial X logical. It anticommutes with some Z logical, and that Z logical has a slab representative in each of the L_x+1 disjoint planes $x=x_0$, so O must place qubits in every one of these planes. To count them, cut the Z slabs of one plane into their pieces: a vertical data column for each input qubit, and the backbone segments. Record, for each piece, whether O meets it an odd number of times. Read a data piece as its input qubit, and a backbone piece as the qubits of its input X check that lie behind it, the first t of them for the tth segment. The records then spell out a nontrivial X logical of the input code, which has at least d_X^{in} entries. The first t qubits of a check may be exchanged for the remaining ones, because the whole check is a stabilizer of the input code, so a backbone piece stands for at most $\eta_X = \max\{1, \lfloor w_X/2 \rfloor\}$ qubits, where w_X is the largest input X-check weight. Each piece met an odd number of times holds at least one qubit of O, so each plane contains at least $\lceil d_X^{\mathrm{in}}/\eta_X \rceil$ qubits of O, and

$$d_X \geq (L_x+1)\left\lceil \frac{d_X^{\mathrm{in}}}{\eta_X} \right\rceil, \tag{4}$$

dually for Z. For balanced good inputs the realization therefore has $n=\Theta(N^3)$, $K=\Theta(N)$, and $d=\Theta(N^2)$, the parameters of Eq. (1) (Supplemental Theorem SI.6). For the $[[4,2,2]]$ and $[[6,4,2]]$ inputs the bound equals the exact distance in every computed instance.

For balanced inputs the block has $\Theta(N)$ heights and $\Theta(N)$ transverse positions, so there are

$$r_X = \Theta(N), \qquad r_Z = \Theta(N) \tag{5}$$

pairwise-disjoint slab regions within each orientation. Each region is a complete port for its logical polarization. This repeated access will also give the structural bounds in Sec. V.

III. LOCAL GAUGING AND UNIVERSAL LOGICAL OPERATIONS

A. Why a three-dimensional grid preserves distance

Gauging measures a logical operator by measuring many small checks whose product is that operator [2]. Let $\overline{P} = X(D)$ be the selected logical representative. Attach each data qubit of D to its own vertex of a connected grid $G=(V,E)$, call these vertices the *ports* U, and place an ancilla qubit on every edge. The new check at a vertex, its *vertex star*, is X on the edges at that vertex and on the attached data qubit, if any. Every edge belongs to two vertex stars, so the product of all vertex stars is $\overline{P}$, and the product of their measured outcomes is the logical outcome; yet each measurement involves only a

few neighboring qubits. The grid also carries Z checks around its unit squares, and every original Z check that meets D is extended along short grid paths joining its ports, so that it commutes with the stars. Preparing the edge qubits in $|0\rangle$, measuring the new checks, and finally reading the edges in Z returns the data to its original code with an updated Pauli frame. The X/Z-exchanged construction measures pure-Z logicals.

The danger is that the added qubits create new, shorter logical operators. The quantity that controls this is the expansion of the grid relative to its ports,

$$\beta(G,U) = \min_{S\subseteq V} \frac{|\delta S|}{\min\{|S\cap U|, |U\setminus S|\}}, \tag{6}$$

where δS is the set of edges leaving S and only positive denominators are included. We call δS a *wall* (not to be confused with the lattice walls of Sec. II): it separates the ports in S from the others, and β is the smallest number of wall edges paid per port separated. For a code with CSS distances d_X, d_Z, the deformed distances satisfy

$$\widetilde{d}_X \geq \min\{1, \beta(G,U)\} d_X, \qquad \widetilde{d}_Z \geq d_Z. \tag{7}$$

The reason is short. An X logical of the deformed code may put part of its weight on the ancilla edges, but those edges must form a wall δS; any other edge set anticommutes with some square Z check. Multiplying by the vertex stars in S removes the wall and puts X instead on the data qubits attached in S, or, using the complement of S, on those attached outside it. Either way the result is a logical of the original code, so it has weight at least d_X. The trade removes $|\delta S|$ edges and, on the better side, adds at most $m = \min\{|S\cap U|, |U\setminus S|\}$ data qubits, while $|\delta S| \geq \beta m$. Hence the deformed logical weighs at least $\min\{1,\beta\}$ times a logical of the original code. If $\beta \geq 1$, no logical saves weight by moving into the ancilla. The Z sector cannot use the ancilla at all: a Z logical reduces to an original Z logical times grid cycles.

The slab geometry makes β a constant, and this is where three dimensions matter. A slab representative lies in one plane and has $\Theta(N^2)$ qubits, hence $\Theta(N^2)$ ports. If the grid were only that plane, a straight line across it would wall off half of the ports with $\Theta(N)$ edges, so $\beta = O(1/N)$. If the grid is a box as thick as it is wide, a wall must be a surface. Either it separates roughly the same region in every layer of the box, and then it cuts through all the layers, or the separated region changes from layer to layer, and every change costs edges between layers. In both cases its area grows in proportion to the number of ports it separates (Supplemental Lemma SIII.8). If U occupies at most t parallel planes of a box with aspect ratio at most A, this gives

$$\beta(G,U) \geq \frac{1}{2At}. \tag{8}$$

Constant refinement assigns distinct local port labels to distinct grid vertices and increases t only by a constant. Unit-square cycles, bounded pairing paths, and bounded local density keep the check weight, diameter, and incidence constant. Thus one block or two blocks at fixed bounded cell separation admit local logical measurement using $O(n)$ ancillas and retaining distance at least $\beta_0 d$ for a fixed $\beta_0 > 0$. Section III B shows that $\beta_0 = 1$ is attainable. The proofs are in Supplemental Sec. SIII.

B. Keeping the distance throughout the computation

A logical measurement can lower the fault distance, the smallest number of faults that cause an undetected logical error, in three ways. The added qubits can carry a short logical operator; a few wrong measurement outcomes can flip the reported result; and a few faults can exploit the join between two consecutive operations. Figure 3 shows how the construction blocks each of them.

The first danger is real for thin interfaces. Consider a joint measurement $\overline{X}_a\overline{X}_b$ of two blocks. Once it has been measured, $\overline{X}_a\overline{X}_b$ is a stabilizer, so $\overline{X}_a$ and $\overline{X}_b$ act in the same way on the code. Take any wall δT with the ports of block a on one side and those of block b on the other. The product of the vertex stars in T is $\overline{X}_a$ times X on the wall, so $X(\delta T)$ alone is a logical operator equivalent to $\overline{X}_a$ [Fig. 3(a)], and

$$\widetilde{d}_X \leq \min\{|\delta T| :\ T\cap U = U_a\}. \tag{9}$$

For layer-code blocks of side $\Theta(N)$, the distance $d = \Theta(N^2)$ is comparable to the area of a block face. A measurement layer of bounded thickness h meets the plane between the blocks in a strip of $O(hN)$ edges, which is a wall, so it would reduce the distance of every joint measurement to $O(hN)$. Joint measurements cannot be avoided: they implement CNOT, phase teleportation, and resource transfer. Supplemental Table S6 confirms the loss on explicit codes: a single added sheet leaves distance at most L where the undeformed code has L^2. A complete circuit-level simulation on explicit layer-code blocks (Sec. VI and Fig. 5) finds the same effect in the fault distance of the measurement circuit, exactly at two block sizes and, with routed nearest-neighbor extraction, for three input codes, and in its failure rate at four sizes. Small hardware runs show it as well (Sec. VII).

Four ingredients keep the distance. First, the grid fills the participating blocks, so every wall between their ports is a surface with $\Omega(N^2)$ edges [Fig. 3(b)]; this is the constant relative expansion of Eq. (8), which the constant slab thickness makes possible. Second, carrying $m_0 = \lceil 2At \rceil$ parallel ancillas on every grid edge multiplies every wall by m_0, so $\beta \geq 1$ and Eq. (7) gives $\widetilde{d}_X \geq d_X$: each measurement keeps the full static distance at constant ancilla cost (Supplemental Corollary SIII.4). Third, the measurement is repeated in time [Fig. 3(c)]. A single wrong outcome of one vertex star flips the reported product, so the new checks are measured for $R = \Theta(d)$ rounds, and a wrong outcome that is not repeated is caught by

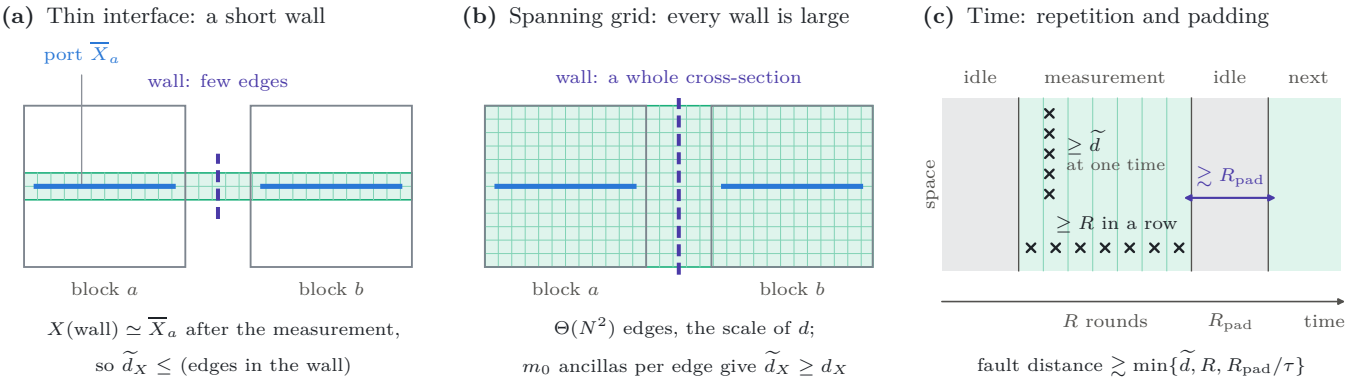


FIG. 3. Why the measurement keeps the fault distance, in a side view of two blocks (squares) with their X slab ports (blue) and the ancilla grid (green). (a) If the grid joins the blocks only through a thin bridge, X on a wall across the bridge (dashed) is equivalent to $\overline{X}_a$ once $\overline{X}_a\overline{X}_b$ is measured, so a few faults cause a logical error. (b) If the grid fills both blocks, every wall between their ports is a whole cross-section, with $\Theta(N^2)$ edges, the same scale as the code distance; m_0 ancillas per edge make every wall at least as heavy as the ports it separates, and the full distance is kept. (c) In time, an undetected error inside a measurement is either a logical at one time step, with at least $\widetilde{d}$ faults, or one outcome wrong in each of the R repeated rounds. Idle rounds between operations force any pattern that links two of them to stretch across the padding.

comparing consecutive rounds. To flip the result undetected, faults must corrupt the same outcome in every one of the R rounds. Williamson and Yoder proved the resulting fault-distance bound for gauging [2]. Because our operations start and end in open idle rounds rather than at perfect boundaries, we give a short direct proof (Supplemental Proposition SV.17). Split the faults by type. Faults of the measured type (X) leave, at the end, an error on the data and a set of flipped edge readouts; together these form one operator of the deformed code, which is a nontrivial logical whenever they cause an error. Every qubit of it carries at least one fault, so there are at least $\widetilde{d}_X$ faults. Faults of the other type either leave a logical error on the data, again with at least d_Z faults, or flip the reported outcome. In that case either every one of the R rounds contains a wrong outcome, or a round without one shows that the errors accumulated so far form a logical operator that anticommutes with the measured one, with at least d_Z faults. Hence the measurement has fault distance at least $\min\{\widetilde{d}_X, d_Z, R\}$, and exact minimization on small instances shows that this bound is attained. Fourth, successive measurements on a block are separated by $\Theta(d)$ idle rounds, so faults that straddle two operations also cost $\Omega(d)$ (Sec. IV).

Resource transfer is the one place where blocks of different distance meet. Joining a small block to a large one cannot expose the large block: the same wall argument, applied to everything except the small block's own logical qubits, shows that every other logical operator keeps the distance of the larger block (Supplemental Lemma SIII.6). The small courier blocks below therefore carry only one traveling resource each, and their errors are charged to distillation. Supplemental Table S9 lists the resulting bound for every operation: each logical qubit that is read later, other than a resource in transit, is protected at every step by fault distance $\Omega(d)$.

C. Gates and encoded resources

Pure-X/Z measurements suffice for the logical gate set. A CNOT uses an ancilla a in $|+\rangle$ and the measurements

$$Z_cZ_a\ (u), \qquad X_aX_t\ (v), \qquad Z_a\ (w), \tag{10}$$

followed by the Pauli-frame update $Z_c^vX_t^{u\oplus w}$. Outcome bit zero denotes eigenvalue $+1$.

For phase teleportation, prepare a in $R_\theta|+\rangle$, where $R_\theta = \mathrm{diag}(1, e^{i\theta})$, measure Z_dZ_a with bit m and X_d with bit n, and retain a. The correction

$$C_{mn} = R_{2\theta}^mZ_a^nX_a^m \tag{11}$$

produces $R_\theta|\psi\rangle$. For $\theta = \pi/2$ this consumes $|Y\rangle = S|+\rangle$; for $\theta = \pi/4$ it consumes $|A\rangle = T|+\rangle$, with a possible S correction using one additional $|Y\rangle$. Exchanging X and Z gives $S_X = HSH$, and $SS_XS = e^{i\pi/4}H$. These identities generate Clifford+T using the one- and two-block measurements above. Complete branch maps and resource counts appear in Supplemental Sec. SIV.

A physical resource enters at a designated bare port. The raw qubit is teleported into an encoded block by a joint ZZ measurement and an X readout. Its weight-one error class is the intentional resource degree of freedom; all spectator logical classes retain distance $\Omega(d)$ in the quotient by that class. Subsequent transfers use only encoded blocks, with distances growing by fixed factors. We call this sequence an encoded courier. Each courier block is dedicated to one traveling resource; its other logical qubits are prepared in fixed states and never read.

The resource supply uses fixed Steane $7\to1$ and punctured Reed–Muller $15\to1$ factories, implemented by the same logical primitives [25–27]. Resource errors are modeled as local-stochastic defective slots: a defective candidate may be in any state, entangled with other candidates or with the environment. Phase teleportation

with any such resource applies, relative to the ideal rotation, an operator in span$\{I, Z\}$ on the coordinate it feeds. A predetermined recovery therefore returns exactly the ideal output whenever at most one coordinate of a run is defective, so each run always produces an output, and the outputs are again a defective family with

$$q_{r+1} \le C_{\mathrm{dist}} q_r^2. \tag{12}$$

The constant counts pairs among the fifteen principal and fifteen correction slots of a run, $C_{\mathrm{dist}} = \binom{30}{2} = 435$. Disjoint runs use disjoint candidate sets; their errors need not be independent, and no twirl or diagonal error basis is assumed. The raw boundary, encoded transfers, and factories are included in the circuit analysis rather than supplied as ideal external resources.

IV. FAULT TOLERANCE AND RESOURCE SCALING

A. From local distance to a complete circuit

We consider local-stochastic Pauli circuit noise: any specified set F_0 of fault mechanisms at distinct locations occurs with probability at most $p^{|F_0|}$. A round of check measurements has constant depth and uses only nearest-neighbor gates on a finite ancilla schedule. Each circuit fault acts like at most $c_1 = O(1)$ elementary faults of the simpler phenomenological model, in which a fault is a data error or a wrong measurement outcome. We count faults in that model and divide by c_1 at the end.

The argument that the whole computation keeps fault distance $\Omega(d)$ has a simple shape. An undetected logical error needs a fault pattern that no detector notices, that is, no comparison of consecutive outcomes and no entry, exit, preparation, or readout check. Split such a pattern into connected pieces, where two faults are connected when they can trigger a common detector; at least one piece causes the error. A piece that lies inside one operation and the idle rounds around it needs at least $\min\{d, R\}$ faults, by Sec. III B. Successive operations on a block are separated by R_{pad} idle rounds. Two connected faults are at most $\tau = O(1)$ rounds apart, so a piece that reaches from one operation to the next must stretch across the padding and needs about R_{pad}/τ faults. Precisely, with δ_g^{rel} the fault distance of measurement window g and $\delta_{\mathrm{mem}}^{\mathrm{rel}}$ that of idle memory, both defined below, the composed phenomenological distance satisfies

$$f_{\mathrm{ph}} \ge \min\left\{ \min_g \delta_g^{\mathrm{rel}}, \delta_{\mathrm{mem}}^{\mathrm{rel}}, 1 + \left\lfloor \frac{R_{\mathrm{pad}} - \tau}{\tau} \right\rfloor \right\}, \tag{13}$$

with $\delta_g^{\mathrm{rel}} \ge \min\{\beta_0 d, R_g\}$ for a window of R_g rounds and $\delta_{\mathrm{mem}}^{\mathrm{rel}} \ge d$. Choosing R_g and R_{pad} of order d makes every term $\Omega(d)$, and the bounded spread c_1 converts this into a circuit-distance bound.

The technical work lies in making "inside one operation" precise without assuming that each operation starts from a perfect state, since earlier faults leave their traces. We cut the schedule in time at idle rounds and carry three pieces of data across each cut: the syndrome s left by earlier errors and the logical Pauli correction λ accumulated so far, which together describe the quantum state, and the outcomes ρ of the last measurement round, which the next detectors compare against. The records must be carried separately because a wrong outcome in the last round before a cut changes ρ but not the state. With these three variables, adjacent operations compose exactly; a direct simulation of an idle → gauging → idle → gauging → idle schedule on explicit layer-code blocks agrees with the composed map on every single phenomenological fault mechanism, and gluing without ρ fails. Each piece of a fault pattern is then judged by its effect *relative* to the same schedule without it: subtracting the fault-free evolution of the same incoming data removes every dependence on earlier errors. This includes a less obvious one: when the logical representatives at two cuts differ by stabilizers, the propagated frame acquires a term Cs_{in} linear in the incoming syndrome. A piece that lies between two cuts placed in idle padding, with one round free of it before the right cut, leaves no syndrome and no records at either cut, so its relative effect is its true effect and the window bounds above apply to it (Supplemental Lemma SV.14).

Resource errors need one more step, because the courier blocks have smaller distance. An error that a resource already carries is passed along with the other incoming data. A new error first created at encoded courier stage j needs a connected fault pattern of size $\Omega(d_j)$ when that stage's measurement and padding times are $\Theta(d_j)$; the raw injection stage is charged separately, with a constant. Distinct resources are kept far apart in spacetime, so one short connected pattern cannot corrupt two of them. The reason is again the padding: a chain of connected faults can leave the courier blocks of one resource only through its transfer into the factory block, and from there it must stay on that block for the R idle rounds that separate the transfer from every other operation on it, whatever cells it visits afterwards (Supplemental Lemma SV.25). A script builds a complete factory schedule by these rules, with cell reuse and adaptive branches, and checks them and the resulting separation on it. Summing over the stages gives a raw-to-encoded resource error parameter $q_0 = O(\sqrt{p})$ uniformly in the terminal distance, and Eq. (12) then suppresses the resource error quadratically.

Every detector has bounded support, so the graph that connects faults through common detectors has bounded degree, uniformly over the finitely many branches at adaptive steps. The committed decoder chooses a minimum-cardinality fault set consistent with the detector history through a fixed buffer beyond the commitment time. A wrong decision about a protected outcome requires a large connected set of fault locations, a fixed fraction of which are actual faults. Counting such connected sets in a bounded-degree graph therefore yields a positive threshold. This is an information-theoretic decoder; efficient classical decoding is not assumed. Supplemental Sec. SV

gives the local detector, relative-distance, resource, and cluster arguments in full.

B. Architecture theorem

The input family is a balanced good CSS LDPC family with boundedly dense available sizes: every sufficiently large target length has an available member within a fixed multiplicative factor. Fixed-degree Quantum Tanner families provide the existential input used here [28, 29]. Let $A \subset \mathbb{Z}^3$ be the finite allocated logical region, $W = |A|$, and T the logical circuit depth. Allocated working sites are included in W; physical gate ancillas and live factory resources are included in the bound below.

To state the geometric overhead, partition the logical lattice into coarse boxes of side $b(f) = \Theta(f^{1/6})$. Define $B_f(A)$ as the number of boxes that meet A but are not completely filled. This counts partial occupancy at the block-packing scale, including boundaries, holes, and fragmentation.

Theorem IV.1 (Universal fault-tolerant architecture)**.** *Use the explicit layer-code realization of Theorem II.1, a balanced good input family with boundedly dense sizes, and the local measurement, resource, and extraction schedules described above. There are constants $p_{\mathrm{th}}, c, f_{\min} > 0$ such that for $0 < p < p_{\mathrm{th}}$ and target failure $0 < \varepsilon < 1/2$, any finite nearest-neighbor three-dimensional logical circuit can be simulated with*

$$f = \max\left\{ f_{\min}, \left\lceil \frac{c \log(WT/\varepsilon)}{\log(p_{\mathrm{th}}/p)} \right\rceil \right\}, \tag{14}$$

using

$$Q_{\mathrm{phys}} = O\Big(Wf + (B_f(A) + 1) f^{3/2} \Big), \tag{15}$$

$$D_{\mathrm{phys}} = O(T f^{3/2} \Psi), \qquad \Psi = O(f^{\log_2 15}). \tag{16}$$

Persistent data remain in layer-code blocks, and physical operations use nearest-neighbor connectivity. For a connected bounded-aspect logical box,

$$Q_{\mathrm{phys}} = O(Wf + f^{3/2}). \tag{17}$$

The scaling follows directly from the code parameters. Choose $d = \Theta(f)$ and measurement, padding, and commitment buffers of order f. Equation (1) then gives

$$N = \Theta(\sqrt{f}), \qquad K = \Theta(\sqrt{f}), \qquad n = \Theta(f^{3/2}). \tag{18}$$

A full coarse logical box fits into one cell. Counting full and partial boxes requires $O(W/K + B_f(A) + 1)$ data/factory cells. Multiplication by n proves Eq. (15). Neighboring logical sites map to the same cell or adjacent cells, and a bounded coloring schedules inter-cell gates.

For a bounded-aspect logical box, $B_f(A) = O((W/\sqrt{f})^{2/3} + 1)$, so the partial-cell term is absorbed into Eq. (17). Figure 4 shows the crossover at $W \sim \sqrt{f}$. Thin or fragmented regions retain the general correction.

For time, a distilled resource takes $O(f\Psi)$ rounds, where ℓ distillation levels give $\Psi = O(15^\ell)$ and $2^\ell = O(f)$. One logical layer requires $O(K)$ resources per data block, and serial production gives $O(Kf\Psi) = O(f^{3/2}\Psi)$ rounds per layer, that is $O(f^{3/2+\log_2 15}) \subseteq O(f^{5.41})$. This serial schedule is chosen to keep the space bound; parallel factories would trade space for depth. A depth-first factory schedule stores $O(\ell)$ candidates at once, which fits within the reserved logical capacity. The complete location count is polynomial in f times $WT\Psi$; the exponential cluster bound and final-resource union bound are both at most $\varepsilon/2$ for the choice in Eq. (14). The detailed theorem and constant ledger appear in Supplemental Theorem SV.31.

The depth deserves as much attention as the space. Per logical layer the minimal-space schedule takes $O(Kf\Psi) = O(f^{3/2+\log_2 15})$ rounds, against $O(f)$ for lattice surgery. With P factories per cell the space becomes $O(Wf P + f^{3/2})$ and the depth $O(TKf\Psi/P)$, so qubit-time is unchanged: the serial schedule minimizes space, not space–time volume. A concrete estimate with stated constants, explicit integer allocations, and distances chosen from each schedule's complete runtime (Supplemental Sec. SXVI) makes the scales explicit. For $W = 100$ sites, $T = 10^6$ layers, $\varepsilon = 10^{-2}$, $p = 10^{-3}$, and injected resource error 10^{-2}, the always-output distillation covered by the theorem does not converge: Eq. (12) requires an injected error below $1/C_{\mathrm{dist}} \approx 2.3 \times 10^{-3}$, and even the ideal factory, whose error map starts at $105q_c^2$ with the per-coordinate error q_c including the S corrections, requires about 5×10^{-3}. With optimistic postselected distillation, which the theorem does not cover, the serial layer-code schedule needs about 10^9 qubits for two months, against 4×10^5 qubits for three days for a serial surface-code layout and 2×10^8 qubits for 23 s for a throughput-limited one. At these distances a layer-code block holds one to three logical qubits; with the same accounting on both sides it stores a logical more cheaply than a surface-code patch only for $d \gtrsim 1100$ (between about 600 and 2300 across the distance scenarios). The asymptotic advantage is therefore far from the practical regime until these constants, and the distillation protocol, improve.

The same architecture argument applies to any three-dimensional CSS family with the parameter scaling in Eq. (1), bounded local density and aspect ratio, boundedly dense sizes, and uniformly slab-supported pure logicals; the explicit realization supplies all of these. These are substantive geometric assumptions; Supplemental Corollary SV.32 states the transfer precisely.

V. PORT PACKING AND SHALLOW LOGICAL GATES

Slab mobility also constrains how logical information can be spatially accessible. For a region A, let $g_X(A)$ be the dimension of the pure-X logical space supported in A, and define $g_Z(B)$ analogously. The rank $r(A, B)$ of

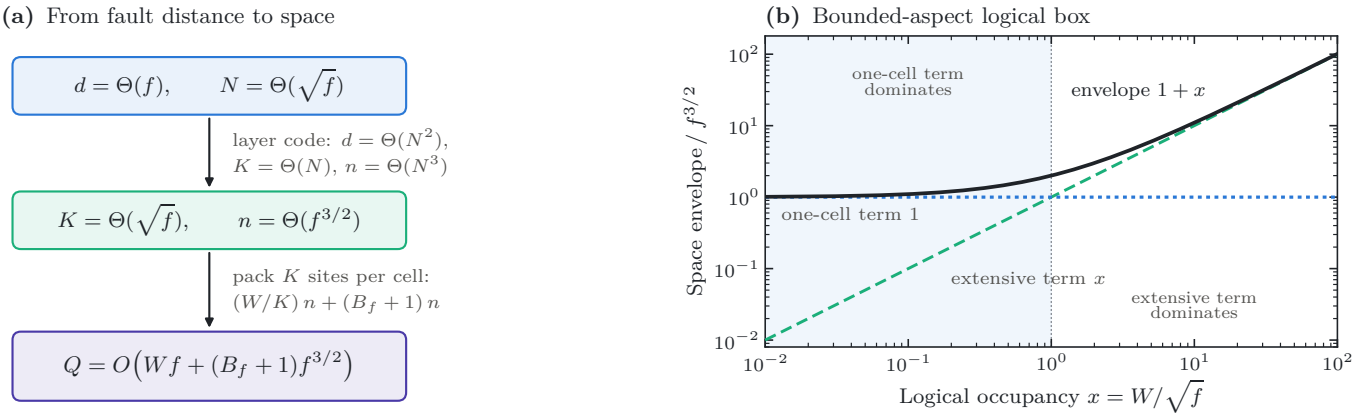


FIG. 4. Origin and crossover of the resource bound. (a) Target fault distance f fixes the cell parameters; packing $K = \Theta(\sqrt{f})$ logical sites per cell yields an extensive Wf term plus the cost of partially occupied cells. (b) For bounded-aspect boxes, the normalized scaling envelope is proportional to $1 + x$, where $x = W/\sqrt{f}$. The one-cell term dominates for $x \ll 1$ and the extensive term for $x \gg 1$. Constants are suppressed: this is an illustration of the asymptotic bound, not a numerical qubit estimate.

their commutation pairing obeys

$$\max\{0, g_X(A) + g_Z(B) - K\} \le r(A, B) \le |A \cap B|. \quad (19)$$

The upper bound follows because all cross commutation factors through the intersection; the lower bound follows from nondegeneracy of the full logical X/Z pairing.

If $A_1, \ldots, A_{r_X}$ are disjoint complete X ports and $B_1, \ldots, B_{r_Z}$ disjoint complete Z ports, summing Eq. (19) over all pairs gives the crossbar bound

$$r_X r_Z K \le n. \quad (20)$$

Every crossing carries rank K, and the crossings are pairwise disjoint. For the balanced layer-code slabs, $r_X, r_Z = \Theta(N)$, $K = \Theta(N)$, and $n = \Theta(N^3)$, so the bound is saturated up to constants. Each transverse intersection has size $\Theta(K)$, while each complete slab has size $\Theta(d)$. Thus the cubic physical size is optimal among CSS families required to provide $\Theta(K)$ disjoint complete ports in both polarizations. This is a bound on simultaneous spatial access, distinct from a storage lower bound.

There is a complementary constraint on unitary processing. Pair the ith X-slab representative and the ith Z-slab representative of any logical Pauli. Each qubit belongs to at most two resulting representatives, giving disjointness $\Delta = \Omega(N)$. The logical-weight bounds sharpen this to

$$\Delta = \Theta(N), \qquad d_\downarrow = \Theta(N^2), \qquad d_\uparrow = \Theta(N^2), \quad (21)$$

where $d_\downarrow$ and $d_\uparrow$ are the smallest and largest minimum weights among nontrivial logical Pauli classes. The Jochym-O'Connor–Kubica–Yoder criterion then forces every fixed-depth, bounded-arity, code-preserving unitary on a sufficiently large static block to be logical Clifford [30]. The result holds even without geometric locality of those gates. The architecture uses adaptive measurements, resource states, and changing codes, which lie outside that unitary setting. Complete packing and disjointness proofs are in Supplemental Secs. SI C and SI D.

VI. FINITE NUMERICAL EVIDENCE

The supplied simulations test finite interfaces in the construction. Three campaigns run complete circuits on explicit layer-code blocks (Fig. 5); the largest campaign, which shows a clean crossing, uses surface codes. That campaign follows a single encoded planar surface code of distance L, with $n = L^2 + (L-1)^2$ data qubits, through an algebraic base-code → gauged-strip → base-code circuit, without resetting the data between stages. The gauged strip is a path of L ancilla qubits attached to a weight-L logical representative, so the campaign tests the gauging interface and the detector bookkeeping of a code switch rather than the layer-code geometry. Its entry and exit satisfy the binary stabilizer transition identities. Noisy repeated extraction produces one detector history, decoded by minimum-weight perfect matching from exact graphlike detector error models.

The SI1000-like model uses two-qubit depolarizing strength p, single-qubit and ordinary idle strength $p/10$, measurement flips $5p$, reset noise $2p$, and data idle-wait noise $2p$. The archived campaign comprises 255 parameter points and 4.534×10^6 shots for sizes $L = 3, 5, 7, 9, 11$ and gauged durations $G = d, 2d, 3d$, where $d = L$ is the code distance. All 15 noiseless controls have zero failures.

For every duration, the failure rate decreases monotonically with L across all five sizes at every tested $p \le 0.004$ (Supplemental Fig. S14). Consecutive-size Wilson intervals are separated at every tested p from 0.0025 to 0.00375 for all three durations. At $p = 0.0035$, increasing L from 3 to 11 reduces the failure rate from 0.1111 to 0.0324 for $G = d$, and from 0.1326 to 0.0428 for $G = 3d$. At $p = 0.0015$ and $G = d$, the reduction is approximately 6.5×10^2.

The smallest-versus-largest crossing medians are 4.471, 4.431, and 4.328×10^{-3} for $G = d, 2d, 3d$, with conditional 95% posterior bands $[4.426, 4.522]$, $[4.392, 4.466]$,

and $[4.265, 4.393] \times 10^{-3}$. Tripling the gauged duration shifts the median by about 3.2%. These are finite-size diagnostics of a surface-code gauging model; they do not estimate the architecture theorem's asymptotic threshold, which concerns layer codes and a committed minimum-weight decoder.

A complete joint measurement on explicit layer-code blocks tests what the surface-code campaigns cannot: the reported bit, both logical sectors, spectator logicals, and an entangled input. Two $[[4, 2, 2]]$-input blocks ($n = 224$, two logical qubits each, distances 5 and 4) measure $\overline{X}_1^{(a)}\overline{X}_1^{(b)}$ by gauging on a planar grid attached to their single-layer slabs, with ancilla-based extraction of every check and uniform circuit noise. Each logical qubit starts maximally entangled with a noiseless reference, and eight observables certify the output Choi state. Without noise every detector and observable is deterministic, so the circuit implements the projective measurement with the correct outcome and leaves the spectators untouched. We then compute, exactly, the minimum number of faults that flips each observable undetectably. With a bridge of $w = 1, 2, 3$ grid rows between the blocks, the fault distance of $\overline{Z}_1^{(a)}\overline{Z}_1^{(b)}$ is exactly w; for $w = 1$ a single fault suffices [Fig. 5(b)]. This is the cut obstruction of Sec. III B at circuit level. A grid spanning both blocks restores it to exactly 5, the blocks' X distance, while the fault distances of the reported bit, set by the number of rounds, and of the spectators do not depend on the interface (Supplemental Sec. SXIV).

A second campaign repeats the measurement on blocks of spacing $s = 2$ to 5 (block distances up to $d_X = 11$, up to 1400 qubits per block), with $2s + 1$ rounds per phase and 3.0×10^6 shots, decoded in each CSS sector separately (Supplemental Sec. SXIV A); on identical shots the full-model BP+LSD decoder of the first experiment (Supplemental Table S17) fails less often through the grid (half as often in the X experiment) and about as often through the bridge, so the contrast below is, if anything, understated. At $p = 10^{-3}$ the failure rate of $\overline{Z}_1^{(a)}\overline{Z}_1^{(b)}$ through the one-row bridge rises from 3.9% at $s = 2$ to 8.8% at $s = 5$, because the undetectable bridge fault has more rounds in which to occur, while through the spanning grid it falls from 8.2×10^{-4} to about 10^{-4} and agrees with the same blocks left idle [Fig. 5(a)]. For the spanning grid each unit of spacing suppresses failures by a factor 4.8 in the X experiment and 2.6 in the Z experiment at $p = 10^{-3}$, falling to one near $p \approx 6 \times 10^{-3}$ [Fig. 5(c) and Supplemental Fig. S16]. At $s = 3$ the fault distance of $\overline{Z}_1^{(a)}\overline{Z}_1^{(b)}$ through a bridge of $w = 1, 2, 3$ rows is still exactly w, although the blocks' X distance is now 7.

A third campaign replaces the colocated ancillas by the routed nearest-neighbor schedule of Supplemental Proposition SV.8, with SWAP-shuttled ancillas, colored conflict classes, and idle noise in every time step, and grows the input at $s = 2$: $[[4, 2, 2]]$, a six-qubit band code, and the Steane code, giving blocks of 224, 720 and 1792 qubits and circuits of up to 16,293 qubits (Supplemental Sec. SXIV B). Without noise all 28 measurement circuits implement the measurement exactly, and the 12 idle controls also give silent detectors and correct observables. Through the one-row bridge the fault distance of $\overline{Z}_1^{(a)}\overline{Z}_1^{(b)}$ is exactly 1 for all three inputs, although the blocks' X distance grows from 5 to at least 18 [Fig. 5(b)]. On the $[[4, 2, 2]]$ blocks with $R = 5$ rounds per phase the spanning grid gives exactly 5, the same as the blocks left idle; through both interfaces the reported bit has exactly $5 = R$, and the spectator X frames have exactly $4 = d_Z$, as in idle memory. Where both are resolved, routing leaves the distances of the colocated circuit unchanged. On the band blocks the reported bit has an undetectable pattern of weight 6 at $R = 7$. The block itself explains this weight: the lightest Z-type operator that anticommutes with the measured logical has weight 6, so no schedule gives the bit more than $\min\{R, 6\}$; whether routing adds lighter patterns is open. At the tested noise strengths the sampled failure rates of these long routed circuits do not distinguish the interfaces, so we use only their exact fault distances.

Complementary memory, local-gauging, factory, and stage-composed tests are reported in Supplemental Sec. SXIII. The deterministic checks cover the complete-code slab certificate, refinement identities, gate maps, 10,008 single-location faults in the finite extraction catalog, and the thin-versus-thick interface comparison of Sec. III B. These checks support the individual interfaces and remain separate from the arbitrary-size proofs.

VII. HARDWARE DEMONSTRATIONS

Three mechanisms of Sec. III B are small enough to run on present superconducting processors: the cut obstruction, its removal by parallel ancillas, and the protection of the reported bit by repeated rounds. We ran them on the 54-qubit IQM Emerald processor through the Open Quantum platform [31] and on the 156-qubit IBM Heron processors `ibm_fez` and `ibm_kingston`, with 6.9 million analyzed samples. The top row of Fig. 6 shows what was run and the bottom row what was measured. The blocks are three-qubit repetition codes, not layer codes, so these runs test the mechanisms and not the architecture. Apart from the routed `ibm_fez` circuits, every circuit was placed without SWAP gates [Fig. 6(a) and Supplemental Fig. S18], and before each submission the exact circuit fault distance of the circuit as sent was computed. A script computes every number below from the archived raw outputs (Supplemental Sec. SXV).

a. Cut obstruction. Two blocks with Z checks Z_0Z_1, Z_1Z_2 and $\overline{X} = X_0X_1X_2$ start in $|000\rangle$ and $\overline{X}_a\overline{X}_b$ is measured by gauging on a graph whose bridge between the blocks has w edges, so that X on the bridge is a logical of weight w [Eq. (9)]; the deformed $\overline{Z}_a\overline{Z}_b$ must stay $+1$. The postselected error falls 35-fold, from 2.4% at $w = 1$ to 0.070% at $w = 2$, on IQM (pooled over three

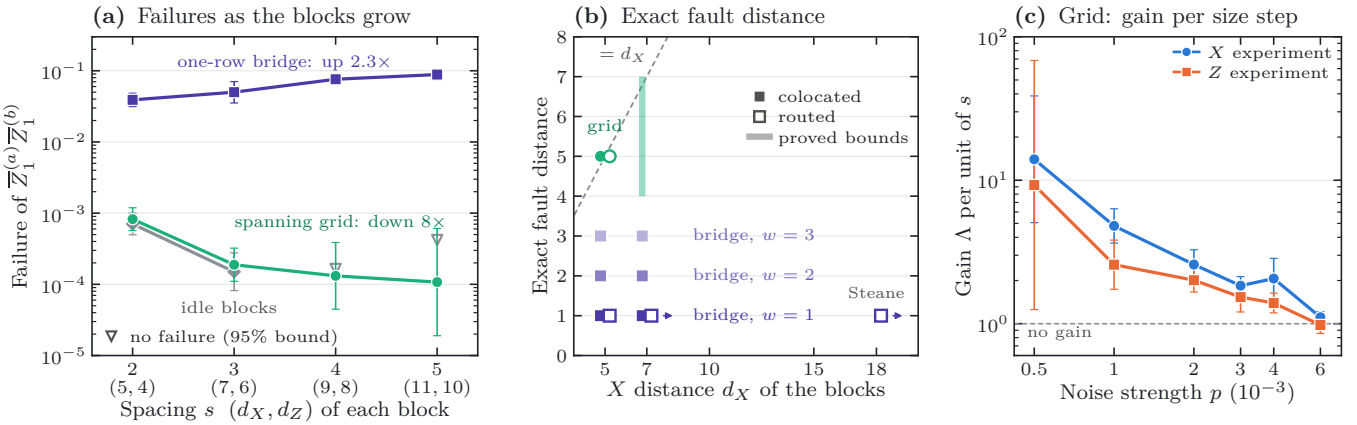


FIG. 5. Two-block logical measurement on explicit layer-code blocks: the cut obstruction at scale, in simulation (the hardware counterpart is Fig. 6). (a) Failure of the measured pair $\overline{Z}_1^{(a)}\overline{Z}_1^{(b)}$ at $p = 10^{-3}$ as the block spacing s grows, through a one-row bridge (violet) and a spanning grid (aqua), and for the same blocks left idle (grey), with Wilson 95% intervals (open triangles: Wilson upper ends of zero counts). (b) Exact circuit fault distance of the measured pair against the X distance of the blocks, computed by CP-SAT. Filled: colocated ancillas at $s = 2, 3$; open: routed nearest-neighbor extraction at $s = 2$ with the $[[4, 2, 2]]$, band and Steane inputs, whose blocks have X distance 5, at least 7 and at least 18 (arrows); bar: the proved lower bound and the weight of an undetectable pattern where the solver stopped. Through a bridge of w rows the distance stays at w while d_X grows; through the grid it equals d_X (dashed) where it is resolved. (c) Factor Λ by which each unit of spacing suppresses failures of any observable through the spanning grid, from a weighted fit over $s = 2$ to 5, with 95% intervals.

runs on two sets of qubits; $p \leq 9.5 \times 10^{-19}$ in each of the two runs with both widths), and from 5.8% to 0.59% and 0.14% at $w = 1, 2, 3$ on IBM, by factors of 9.7 and 4.3 [Fig. 6(a,d)]. When the cycle checks that the extra bridge edges create are ignored, the error returns to the width-one level (2.2 to 3.3% on IQM, 7.2 to 7.9% on IBM), so the gain comes from the larger wall and not from discarding more samples. On IQM, width three had no failure in 11,529 kept samples, but its difference from width two is not significant ($p = 0.071$, in a test fixed before the run).

b. Parallel ancillas. Two parallel ancillas on a single bridge edge ($m_0 = 2$) give 0.53% instead of 2.4%, a 4.6-fold gain ($p = 5.1 \times 10^{-10}$), and 2.3% when the new two-edge check is ignored [Fig. 6(b,e)]. The first CNOT order we tried had circuit fault distance one: a fault on the vertex ancilla between its two bridge CNOTs put X on both parallel edges, which the two-edge check cannot see. Separating the two CNOTs restores distance two. This is the constant-factor loss from circuit faults that Sec. IV allows for; a suitable CNOT order avoids it.

c. Repeated rounds and errors that survive a reset. For the reported bit we use phase-flip blocks ($d_Z = 3$) joined by one edge and measure the two vertex stars and the four block checks in R rounds [Fig. 6(c)]. With fresh ancillas in each round (IQM, which does not allow a measured qubit to be reused), the error falls 20-fold, from 6.2% at $R = 1$ to 0.31% at $R = 2$, close to the model's 6.9% and 0.34% ($p = 2.4 \times 10^{-188}$, fixed in advance). On IBM, where the ancillas are measured and reset between rounds, it falls from 4.6% to 1.1% and 0.59% for $R = 1, 2, 3$ (both steps $p < 10^{-300}$, fixed in advance), but far less than the calibrated model predicts (0.18% and 0.0086% at $R = 2, 3$); at $R = 3$ the error is 68 times the prediction [Fig. 6(f)]. The cause is an ancilla error that survives the reset: a check ancilla reads 1 in two consecutive rounds, with no data error to explain it, in 1.15% of samples, 23 times the rate that independent errors give, and a second reset removes only 3% of these events ($p = 8.8 \times 10^{-6}$) [Fig. 6(g)]. Such a fault, most likely leakage, corrupts the same outcome in every round. It lies outside the local-stochastic model of Sec. IV, which charges every round separately.

d. What the runs show. Together, the runs reproduce at small scale the three facts that the distance argument of Sec. III B combines: the smallest wall bounds the distance of a joint measurement (as Fig. 5 shows in simulation on layer-code blocks), parallel ancillas widen every wall that crosses their edge, and repeated rounds protect the reported bit. They also show which assumption of the theorem present hardware does not meet: faults that persist across rounds. They do not test layer codes, the grid expansion, the threshold, or the resource scaling, which need far larger devices.

VIII. DISCUSSION

The main architectural contribution is a local route from layer-code storage to universal processing. Prefix-parity repair exposes logical operators in single-layer slabs of an explicit realization; grid gauging turns those slabs

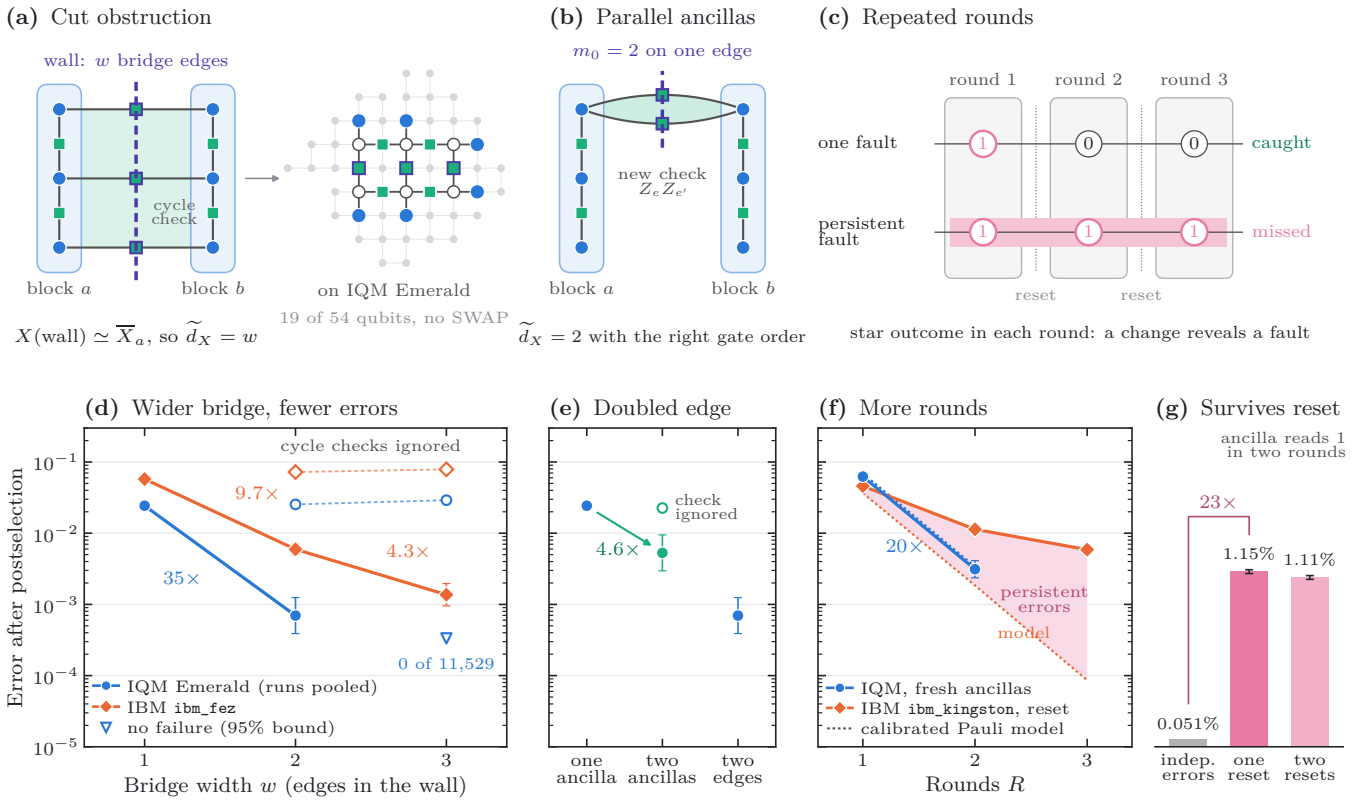


FIG. 6. Hardware demonstrations of the three distance mechanisms. Top row: what was run (schematics). (a) Two three-qubit blocks (blue) joined by a bridge of w edges, each with an edge ancilla (aqua squares); X on the wall of bridge edges (violet) is equivalent to $\overline{X}_a$, so the deformed X distance and the exact circuit fault distance are both w. Shaded: the cycle checks that the extra bridge edges create. Right: the circuit for $w = 3$ as placed on IQM Emerald, with the couplers it uses. (b) Two parallel ancillas on one bridge edge ($m_0 = 2$) add the two-edge check $Z_e Z_{e'}$. (c) A wrong star outcome in one round is revealed by the change in the next round; a fault that persists across rounds repeats the same wrong outcome and is missed. Bottom row: what was measured, as the error after postselection on every detector, with Wilson 95% intervals (open triangle: Wilson upper end of a zero count). (d) IQM runs pooled; open symbols: the same samples with the cycle checks ignored. (e) IQM, one bridge edge with one or two ancillas, and two bridge edges; open symbol: the two-edge check ignored. (f) Reported-bit error against the number of rounds R, with the calibrated Pauli models (dotted); IQM uses fresh ancillas in every round, `ibm_kingston` measures and resets them. Shaded: the excess over the model (run A). (g) Rate at which a check ancilla on `ibm_kingston` (run B) reads 1 in two consecutive rounds with no data error to explain it, against the rate that independent errors give, with one or two resets between rounds. Every run separately: Supplemental Fig. S17.

into protected measurements; and a composed detector schedule supports resource transfer and distillation in the same geometry. Because a layer-code distance is comparable to the area of a block face, every joint measurement must cross a constant fraction of that face; a thick grid does so and, with a constant ancilla multiplicity, keeps the full distance, whereas a thin measurement layer cannot. Simulations on layer-code blocks and small hardware runs show this contrast (Figs. 5 and 6). For bounded-aspect logical regions, the resulting space bound separates an extensive Wf cost from the minimum $f^{3/2}$ cost of one encoded cell.

The port-packing result gives a second interpretation of the layer-code parameters. Their cubic physical volume supports linearly many disjoint complete ports of each polarization, saturating the crossbar bound. Growing mobility also explains why adaptive measurement and resource injection are natural tools: shallow unitaries preserving a static large block are restricted to the Clifford group.

The next steps are a practical decoder for deformation histories, with changing checks, inherited frames and records, and joint measurements (the layer-code decoders of Ref. [32] are a natural starting point, although their memory guarantees do not yet cover code switching); routed circuit-level benchmarks on inputs that grow into the balanced regime, with a shorter routed schedule and a decoder close to minimum weight; bounds on how many logical measurements can share the slab interfaces, compared with parallel code surgery [3] under three-dimensional locality; and factories whose constants make the space–time estimate competitive, possibly in a

different code with a protected local transfer. Random layer codes built from random CSS inputs [32] are a natural candidate for reducing those constants; the explicit realization applies to any CSS input, but the resource accounting would need their parameter scaling. The present theorem uses minimum-weight committed decoding and local-stochastic Pauli noise. Extensions to coherent or leakage noise require an additional fault-path analysis; the IBM runs of Sec. VII show that ancilla errors that persist across rounds already limit the benefit of repeated rounds on present hardware, so leakage reduction would belong in any implementation of the extraction schedule. Deterministic selection of the input component codes is a separate constructive problem. The storage benchmark in Supplemental Sec. SVI makes the restricted regime for comparison with an extensive lower bound explicit, including the condition $W = \Omega(\sqrt{f})$ when the one-cell term is to be subleading.

DATA AVAILABILITY

The simulation datasets, finite algebraic certificates, verification scripts, and figure-generation code are included in the accompanying source and reproducibility package. The Supplemental Material describes the noise models, decoders, stopping rules, and scope of each dataset. The hardware circuits, job identifiers, raw outcomes, calibration snapshots, and analysis scripts are in the folder `hardware` of the package.

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